ic_autolab round 21 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:24:00Z at git `f7fed8638a0b83af3e221e89a0b9398fa3889d50`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-fe8236` (`python3 tools/pcc/pcc.py standing PCC-20261104-fe8236`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**no_challenger**	none (best candidate `wsum-b8-n3`, train S 9.467 / 9.541 / 10.206)	-	-	-	-	-	5.489 / 5.735 / 7.951	1.470 / 1.573 / 2.505	inc: 3.7x / 3.6x / 3.2x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

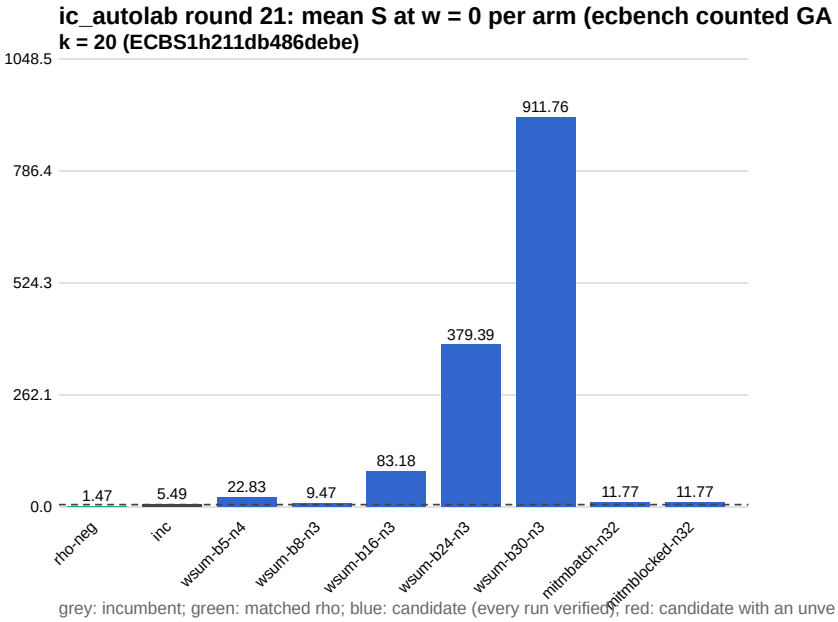
- k = 20: no_challenger

Train (development screen, in sample)

k = 20: session `ECBS1h211db486debe` (run exit 0, verify exit 0, receipt sha256 `88f5aca8734bd1a51e57390c11d4d21bdd9fe86c9e089381d19c3254b8c855ee`)

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.470 / 1.573 / 2.505	matched reference
`inc`	wsum-b5	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=5,free=1`	32	32	5.489 / 5.735 / 7.951	incumbent
`wsum-b5-n4`	wsum-b5	4	`prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=5,free=1`	32	32	22.826 / 22.883 / 23.393	incumbent neighbourhood: N = 1.2 x 3
`wsum-b8-n3`	wsum-b8	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=8,free=1`	32	32	9.467 / 9.541 / 10.206	cross-size transfer: wsum-b8-n3 is the k = 24 incumbent at N / N_a = 0.037
`wsum-b16-n3`	wsum-b16	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=16,free=1`	32	32	83.185 / 86.489 / 116.227	exploration: family wsum-b16 untried at k = 20, N = 0.0001 x N_a
`wsum-b24-n3`	wsum-b24	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=24,free=1`	32	32	379.385 / 406.458 / 650.112	exploration: family wsum-b24 untried at k

arm	family	N factor base	oracle	runs verified		S (w = 0 / 0.1 / 1)	reason proposed
							= 20, N = 0.0001 x N_a
`wsum-b30-n3`	wsum-b30	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=30,free=1`	32	32	911.758 / 1000.782 / 1801.998	exploration: family wsum-b30 untried at k = 20, N = 0.0001 x N_a
`mitmbatch-n32`	mitmbatch	32	`prime-projected-abscissa:size=32,scan_cap=512`	`mitm-prime-batch:m=2`	32	32	11.769 / 12.631 / 20.389	exploration: family mitmbatch untried at k = 20, N = 1 x N_a
`mitmblocked-n32`	mitmblocked	32	`prime-projected-abscissa:size=32,scan_cap=512`	`mitm-prime-blocked:m=2`	32	32	11.769 / 12.631 / 20.389	exploration: family mitmblocked untried at k = 20, N = 1 x N_a



train screen

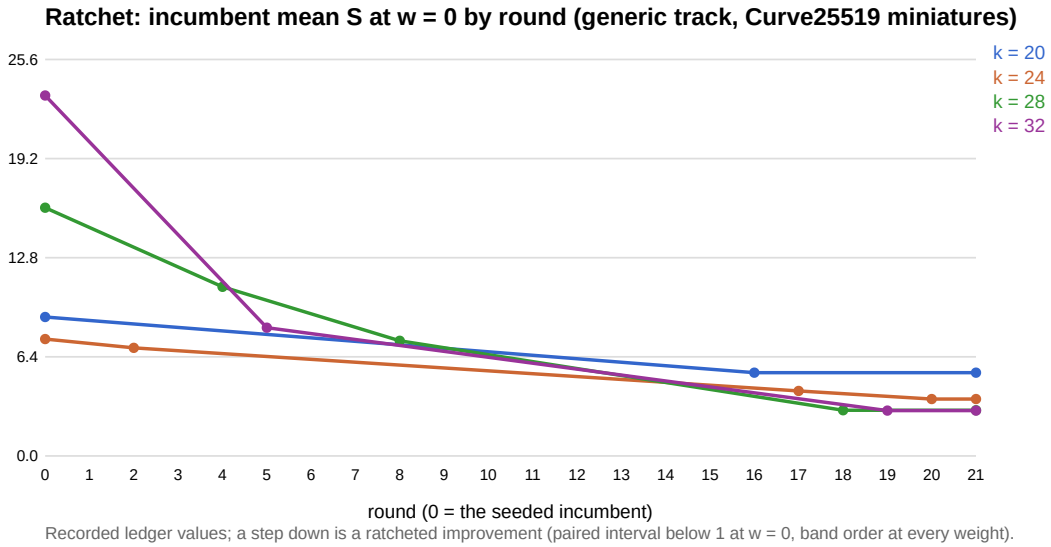
Selection

- k = 20: no_challenger (Pareto alternative at w = 1: `wsum-b8-n3`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	`wsum-b8-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=8,free=1`)	20	3.678 / 3.920 / 6.100	3.0x / 3.0x / 2.7x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
28	`wsum-b12-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=12,free=1`)	18	2.949 / 3.222 / 5.676	3.0x / 3.0x / 3.0x
32	`wsum-b20-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=20,free=1`)	19	2.937 / 3.213 / 5.697	3.1x / 3.1x / 3.1x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.